

ChatGPT Prompts for Data Analyst

A Systematic Approach to AI-Assisted Analysis

This guide provides a systematic approach to data analysis using AI assistance. Following these prompts in sequence will help you thoroughly understand your datasets, identify meaningful insights, and conduct rigorous analysis. Each step builds upon the previous one, creating a comprehensive analytical framework.

Step 1: Data Quality Assessment

The foundation of any analysis begins with understanding what data you have and whether it's reliable.

PROMPT 1: INITIAL COLUMN EXPLORATION

List all the columns in the attached spreadsheet and show me a sample of data from each column.

Purpose: Get a bird's-eye view of your dataset structure and content types.

PROMPT 2: EXPANDED SAMPLING

Take 5 more random samples of the data for each column to make sure you understand the format and type of information in each column.

Purpose: Validate your initial understanding and catch variations in data format that might not appear in a single sample.

PROMPT 3: DATA QUALITY AUDIT

Run a data quality check on each column. Specifically look for: 1. Missing, null, or empty values (give me counts and percentages) 2. Unexpected formats or data types 3. Outliers or suspicious values

Purpose: Identify data quality issues that could compromise your analysis or require cleaning.

Step 2: Introspection & Question Discovery

Once you understand your data structure, explore what stories it can tell.

PROMPT 1: QUESTION GENERATION

Tell me 10 interesting questions we could answer with this dataset and explain why each would be valuable.

Purpose: Brainstorm analytical possibilities and identify high-value questions aligned with potential business needs.

PROMPT 2: FEASIBILITY ASSESSMENT

For [these questions], tell me exactly which columns you'd need to use and whether the current data is sufficient to answer it.

Purpose: Validate whether your desired analysis is possible with available data and identify any gaps.

PROMPT 3: LIMITATION IDENTIFICATION

What questions do you think someone would WANT to ask about this data but we CAN'T answer due to missing information?

Purpose: Understand the boundaries of your analysis and identify opportunities for future data collection.

Step 3: Goal-Oriented Analysis

With a clear understanding of your data and its possibilities, focus your analysis on specific objectives.

PROMPT: STRATEGIC FOCUS

My goal is to [specify your goal]. Given this goal, which aspects of the data should we focus on?

Purpose: Prioritize analytical efforts on the most relevant data dimensions for your specific objective.

Examples of goals you can use:

- "My goal is to find the best/top 1000 customers, I should target for marketing my new products."
- "My goal is to reduce customer churn by 15%."
- "My goal is to optimize marketing spend across channels."
- "My goal is to identify our most profitable customer segments."

Best Practices & Pitfalls

Best Practices

- Follow the Sequence:** Skipping steps may result in incomplete understanding or flawed analysis.
- Iterate When Necessary:** Don't hesitate to circle back to earlier steps if you discover new information.
- Customize for Your Context:** Adapt these prompts to your specific industry or objectives.
- Document Your Findings:** Keep a record of responses for reproducibility.
- Validate AI Outputs:** Always verify insights against domain knowledge.
- Combine with Human Judgment:** Use AI as an assistant, not a replacement for expertise.

Common Pitfalls to Avoid

- Rushing to Analysis:** Skipping data quality checks leads to garbage-in, garbage-out.
- Ignoring Limitations:** Failing to acknowledge what you can't answer is dangerous.
- Over-reliance on AI:** Strategic thinking remains a human responsibility.
- Forgetting Your Audience:** Always consider who will consume your analysis.

Advanced Applications

FOR TIME SERIES DATA

Identify any seasonality, trends, or anomalies in the time-based columns of this dataset.

FOR SEGMENTATION ANALYSIS

Based on the available data, suggest meaningful ways to segment [customers/products/regions] and explain the potential value of each segmentation approach.

FOR PREDICTIVE MODELING

Which variables in this dataset would be most predictive of [target outcome]? Explain your reasoning and identify any potential multicollinearity concerns.

Conclusion

Mastering these prompts will dramatically accelerate your data analysis workflow while maintaining analytical rigor. The key is to use AI as an enhancement to, not a replacement for, critical thinking and domain expertise. Remember: The quality of your analysis depends not just on the tools you use, but on the questions you ask.

Happy Analyzing!

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